Electromagnetic junction conditions in general relativity

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Abstract

This paper derives the electromagnetic (EM) junction conditions in general relativity using distributional approach. We derive the EM junction conditions for both null and non-null case.

1 Introduction

In general relativity, junction conditions are boundary conditions at spacetime boundaries that arise from joining two distinct spacetimes. These spacetime boundaries are described by a hypersurface Σ that, in the most general cases, has different signatures at different points [1]. The most studied junction conditions are those of the metric and its derivatives, as the metric defines the spacetime geometry. For a simple introduction to the metric junction conditions in general relativity, see [2].

Junction conditions in general relativity can include all types of fields besides the metric. These include, for example, scalar field junction conditions in the scalar-tensor theories [3], and gravitoelectromagnetic junction conditions [4]. The metric junction conditions are also used in brane world cosmology, where our universe is a hypersurface (brane) in a higher dimensional spacetime (bulk) [5]. In these theories, gravity in the bulk can induce fields such as an electromagnetic field in the brane [6].

In this paper, we consider junction conditions for classical electromagnetic (EM) fields in the context of general relativity, generalizing the jump conditions of classical electromagnetism. We derive these junction conditions using distributional methods, which are commonly used in modern derivations of the Israel junction conditions [7].

For the EM junction conditions the work by Rawson-Harris [8], where the covariant electromagnetic boundary conditions (junction conditions) are derived for non-null hypersurfaces, is the most relevant. In this paper we generalize this to include dynamic fields and the case where the boundary is null, and do so completely in quantities intrinsic to the hypersurface. Another relevant work is that of [9], which proves that in any variational field theory it is possible to have a well-defined junction conditions. It also provides a general framework to compute junction conditions in any reasonable classical field theory. The results from applying this general framework leads to same junction conditions for electromagnetic fields as our derivation.

In Section 2 we briefly introduce the geometry of the problem and thin shell formalism. The electromagnetic junction conditions for non-null boundaries are derived in Section 3 using the Maxwell equations in curved spacetime. In Section 4 we derive them for the null boundary. In section 5 we calculate both the metric and the EM junction conditions for a static spherically symmetric star of constant density with a surface charge density such that the exterior spacetime will be Reissner-Nordström. We conclude the paper in Section 5 with a discussion of our results.

2 Geometry

In this paper, we work in (3+1)-dimensional spacetime with the metric signature $(-,+,+,+)$. Let $\mathcal{V}$ be a spacetime region containing two distinct spacetime regions $\mathcal{V}^+$ and $\mathcal{V}^-$ separated by a hypersurface Σ. In $\mathcal{V}^\pm$ we have distinct coordinates $x^{\alpha\pm}$, but we assume that we can use coordinate x^α, that is distinct from $x^{\alpha\pm}$, such that it overlaps with $x^{\alpha\pm}$ in open region around Σ.

A hypersurface Σ can be described by a function

$$\Phi(x^\alpha) = 0\,, \tag{1}$$

or by a parametric equation

$$x^\alpha = x^\alpha(y^a), \tag{2}$$

with y^a being intrinsic coordinates of Σ [7]. (Throughout this paper, we use Latin indices to refer to the 3D coordinates of Σ, and Greek indices to refer to the 4D coordinates in $\mathcal{V}$.) Σ is characterized by its holonomic basis vectors, the normal vector, the induced metric, and the extrinsic curvature. The holonomic basis vectors are

$$e^\alpha_a = \frac{\partial x^\alpha}{\partial y^a}\,. \tag{3}$$

The normal vector is defined as [7]

$$n^\alpha = \frac{\varepsilon \Phi_{,\alpha}}{|g^{\mu\nu}\Phi_{,\mu}\Phi_{,\nu}|}\,, \tag{4}$$

where

$$n^\alpha n_\alpha = \varepsilon \equiv \begin{cases} -1 & \text{if } \Sigma \text{ is spacelike,} \\ +1 & \text{if } \Sigma \text{ is timelike.} \end{cases} \tag{5}$$

The normal vector is defined such that it points towards the direction of increasing Φ such that $n^\alpha \Phi_{,\alpha} > 0$. Usually we can figure n^α from the conditions $n^\alpha n_\alpha = \varepsilon$ and $n^\alpha u_\alpha = 0$, where u^α is four-velocity of an observer co-moving with Σ. For stationary Σ, $n^\alpha u_\alpha = 0$ is trivial, in which case $n^\alpha n_\alpha = \varepsilon$ is enough to determine n^α. The induced metric of Σ is

$$h_{ab} = g_{\alpha\beta} e^\alpha_a e^\beta_b\,, \tag{6}$$

and the extrinsic curvature is

$$K_{ab} = n_{\alpha;\beta} e^{\alpha}_{a} e^{\beta}_{b} \,. \tag{7}$$

The extrinsic curvature can also be defined in terms of the Lie derivative of the metric along the normal n:

$$K_{ab} = \frac{1}{2} (\pounds_n g_{\alpha\beta}) e^{\alpha}_{a} e^{\beta}_{b} \,. \tag{8}$$

The induced metric and the extrinsic curvature are important for formulating the junction conditions for the metric field in a coordinate-independent way.

To complete the geometric picture we introduce a parameter l that denotes either proper distance or proper time for non-null Σ that describe the congruence of geodesics that intersect Σ at $l = 0$. Thus, a displacement away from the hypersurface along one of the geodesics is described by $dx^{\alpha} = n^{\alpha} dl$, where n^{α} is the normal vector.

2.1 Null geometry

When the hypersurface separating the spacetimes is null (lightlike), the geometry changes quite a lot. A null manifold has a degenerate metric and can be described by a Carrollian geometry.

Since the metric is degenerate the normal vector defined by

$$k_{\alpha} = -\Phi_{,\alpha}, \tag{9}$$

is a null vector, and tangent to the null hypersurface and the null generators [7]. For the null case it is useful to introduce the coordinates based on the null generators:

$$y^{a} = \{\lambda, \theta^{A}\}, \tag{10}$$

where $A = 2, 3$. The coordinates θ^{A} label the null generators. The coordinate λ parametrises the generator such that displacement along each null generator is $dx^{\alpha} = k^{\alpha} d\lambda$. Since the null normal is also a tangent we can define as the null normal as

$$k_{\alpha} \equiv e^{\alpha}_{\lambda} = \frac{\partial x^{\alpha}}{\partial \lambda}. \tag{11}$$

To complete the basis we need the null transversal vector N^{α} defined by [7]

$$N_{\alpha} k^{\alpha} = -1 \quad \& \quad N_{\alpha} e^{\alpha}_{A} = 0. \tag{12}$$

The induced metric (non-degenerate metric in the two-dimensional subsurface of Σ) is

$$\sigma_{AB} = g_{\alpha} e^{\alpha}_{A} e^{\beta}_{B}. \tag{13}$$

For the null hypersurface instead of extrinsic curvature, we define transverse curvature that carries information about the derivative of the metric in the transverse direction [7]

$$C_{ab} \equiv -N_{\alpha} e^{\alpha}_{a;\beta} e^{\beta}_{b} \tag{14}$$

$$= \frac{1}{2} \left(\pounds_N g_{\alpha\beta} e^{\alpha}_{a} e^{\beta}_{b} \right) \tag{15}$$

To complete the geometric picture for null case we introduce a parameter τ such that it intersects Σ at $\tau = 0$. For the null case we cannot adapt a unique congruence, so we use an arbitrary congruence of timelike geodesics intersecting Σ[7]. We introduce arbitrary vector field $u^\alpha_\pm$, such that displacement along a congruence is

$$dx^\alpha = u^\alpha d\tau. \tag{16}$$

This vector field $u^\alpha_\pm$ can be associated with the four-velocity of a family of observers making measurements on the shell[7]. To ensure that the congruence is smooth at the hypersurface we demand $[u^\alpha] = 0$.

2.2 Israel junction conditions and the thin shell formalism

Israel junction conditions are the junction conditions for the metric field when the boundary between two spacetimes is a non-null regular hypersurface Σ. The first and the second junction conditions are [7, 10]

$$[h_{ab}] = 0\,, \qquad\qquad [K_{ab}] = 0\,. \tag{17}$$

The square brackets denote the difference in quantities across Σ:

$$[A] \equiv A^+\big|_\Sigma - A^-\big|_\Sigma\,, \tag{18}$$

the labels for spacetime regions are such that n^α points towards $\mathcal{V}^+$.

The second junction condition $[K_{ab}] = 0$ is violated if there is a singular (thin) layer of matter at Σ. This leads to Σ having a stress tensor according to Lanczos equation:

$$S_{ab} = -\frac{\varepsilon}{8\pi}([K_{ab}] - [K]h_{ab}), \tag{19}$$

where $K = h^{ab}K_{ab}$ [2, 7]. This model with the second junction condition violated is usually called the thin shell formalism.

For the null case the junction conditions are analogous to the non null case [7, 2]

$$[\sigma_{AB}] = 0 \quad \& \quad [C_{ab}] = 0. \tag{20}$$

As for the when there is a thin shell the surface quantities of the shell are given by [7, 2]

$$\mu = -\frac{1}{8\pi}\sigma^{AB}[C_{AB}] \tag{21}$$

$$j^A = \frac{1}{8\pi}\sigma^{AB}[C_{\lambda B}] \tag{22}$$

$$p = -\frac{1}{8\pi}[C_{\lambda\lambda}], \tag{23}$$

where μ is the surface density of the shell, j^A is the surface current and p is the pressure.

Now that we have a basic understanding of the geometry of hypersurfaces and the junction conditions, we are ready to derive the electromagnetic junction conditions.

3 EM Junction conditions for non-null hypersurfaces

In this section we work with the geometry of spacetime described in the previous section. We have a non-null hypersurface Σ separating two spacetimes regions $\mathcal{V}^+$ and $\mathcal{V}^-$, where the Maxwell's equations apply. Maxwell's equations in curved spacetimes are

$$F^{\alpha\beta}{}_{;\beta} = 4\pi J^\alpha \,, \tag{24}$$

$$F_{[\alpha\beta;\gamma]} \equiv F_{\alpha\beta;\gamma} + F_{\beta\gamma;\alpha} + F_{\gamma\alpha;\beta} = 0 \,, \tag{25}$$

where $F^{\alpha\beta}$ is the electromagnetic tensor defined via the vector potential A^μ [11]:

$$F_{\alpha\beta} = A_{\beta;\alpha} - A_{\alpha;\beta} \,. \tag{26}$$

We are using the convention of Misner, Thorne, and Wheeler where the electric and magnetic fields are defined by

$$F^{0j} = -F^{j0} = E^j \,, \qquad\qquad F^{jk} = \epsilon^{jkl} B^l \,, \tag{27}$$

where ϵ^{jkl} is the Levi-Civita symbol [11].

We formulate the EM junction conditions by investigating the necessary conditions for obtaining valid distributional solutions to the Maxwell's equations. For the solutions to be valid they also need to produce a distributionally valid energy-momentum tensor for electromagnetism, which is defined by [11]

$$T^{(EM)}{}_{\alpha\beta} = \frac{1}{4\pi}\left(F_{\alpha\mu} F_\beta{}^\mu - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} g_{\alpha\beta} \right) . \tag{28}$$

3.1 First condition

We start with a distributional vector potential as the solution to the Maxwell's equations

$$A_\mu = \Theta(l) A^+{}_\mu + \Theta(-l) A^-{}_\mu \,. \tag{29}$$

Using the definition for the electromagnetic tensor (26) we get

$$F_{\alpha\beta} = \Theta(l) F^+{}_{\alpha\beta} + \Theta(-l) F^-{}_{\alpha\beta} + \varepsilon\delta(l) \left([A_\beta] n_\alpha - [A_\alpha] n_\beta \right) . \tag{30}$$

This will produce a term proportional to $\Theta(l)\delta(l)$ in the energy-momentum tensor, as it is quadratic in $F_{\alpha\beta}$. To avoid this distributionally invalid term, we require the constraint $[A_\beta] n_\alpha - [A_\alpha] n_\beta = 0$. This constraint leads directly to our first EM junction condition:

$$[A_a] = 0 \,. \tag{31}$$

This means that the vector potential must have continuous tangential components while the normal component may have jump across Σ. The direct consequence of this is that the junction condition is automatically expressed in terms of the intrinsic coordinates of Σ.

With the first EM junction condition in place, the distributional electromagnetic tensor simplifies to

$$F_{\alpha\beta} = \Theta(l) F^+{}_{\alpha\beta} + \Theta(-l) F^-{}_{\alpha\beta} \,. \tag{32}$$

Plugging the electromagnetic tensor above into the Maxwell's equation (25) gives us

$$0 = [F_{\alpha\beta}]n_\gamma + [F_{\beta\gamma}]n_\alpha + [F_{\gamma\alpha}]n_\beta \,, \tag{33}$$

This equation is trivial when two indices are the same, due to antisymmetry. This simplifies to

$$[F_{ab}] = 0 \,. \tag{34}$$

The simplest way to see this is to work in a basis with holonomic basis vectors of Σ and n^α as the basis vectors, where this becomes trivial. This junction condition is not unique since $[A_a = 0]$ directly implies $[F_{ab}] = 0$.

3.2 Second condition

The covariant derivative of the electromagnetic tensor is

$$F^{\alpha\beta}{}_{;\beta} = \Theta(l)F^{+\alpha\beta}{}_{;\beta} + \Theta(-l)F^{-\alpha\beta}{}_{;\beta} + \varepsilon\delta(l)[F^{\alpha\beta}]n_\beta \,. \tag{35}$$

The Maxwell's equation (24) implies

$$J^\alpha = \Theta(l)J^{+\alpha} + \Theta(-l)J^{-\alpha} + \delta(l)I^\alpha, \tag{36}$$

where $I^\alpha = J^\alpha_\Sigma$ is the four-current density on Σ. This means that the surface current density on Σ is given by

$$\varepsilon[F^{\alpha\beta}]n_\beta = 4\pi I^\alpha \quad . \tag{37}$$

This is our second EM junction condition, and we can simplify it in terms of intrinsic coordinates of Σ by using the antisymmetric properties of $F^{\alpha\beta}$.

By contracting (37) with the normal vector n_α and using antisymmetric properties of the electromagnetic tensor we get

$$[F^{\beta\alpha}]n_\alpha n_\beta = -[F^{\beta\alpha}]n_\alpha n_\beta \,. \tag{38}$$

This is only true for $[F^{\alpha\beta}]n_\alpha n_\beta = 0$, meaning surface current doesn't have any component along the normal direction. This reduces the second EM junction condition to

$$\varepsilon[F^a] = 4\pi I^a \,, \tag{39}$$

where $F^a = F^{a\beta}n_\beta$.

3.3 Energy-momentum tensor

As a consequence of the first electromagnetic junction conditions, the electromagnetic energy-momentum tensor has the form

$$T^{(EM)}{}_{\alpha\beta} = \Theta(l)T^+{}_{\alpha\beta} + \Theta(-l)T^-{}_{\alpha\beta} \,, \tag{40}$$

where $T^{\pm}{}_{\alpha\beta} = \frac{1}{4\pi}\left(F^{\pm}{}_{\alpha\mu}F_{\beta}{}^{\pm\mu} - \frac{1}{4}F^{\pm}{}_{\mu\nu}F^{\pm\mu\nu}g^{\pm}{}_{\alpha\beta}\right)$ with the $\pm$ signs describing the quantities in different sides Σ. This result tells us that the energy-momentum tensor produced by an electromagnetic field cannot be singular even if there is charge density at Σ and the electromagnetic tensor $F_{\alpha\beta}$ is discontinuous. This is not surprising since $T^{\alpha\beta}$ is quadratic in $F_{\alpha\beta}$—this restricts $F_{\alpha\beta}$ to be regular at Σ—but it is an interesting contrast to the energy-momentum tensor of matter, which can be singular at Σ.

4 EM junction conditions for null hypersurfaces

In this section we work with the null geometry described in section 3.

4.1 First condition

We start similarly to the non-null case by demanding the potential produce a valid distributional electromagnetic field tensor:

$$k_\alpha[A_\beta] - k_\beta[A_\alpha] = 0 \tag{41}$$

Contracting this with the basis vectors gives the following conditions:

$$e^\alpha_A: \quad [A_A] = 0, \tag{42}$$
$$k^\alpha: \quad [A_\beta k^\beta] = 0. \tag{43}$$
$$N^\alpha: \quad [A_\beta] = -[A_\alpha N^\alpha]k_\beta. \tag{44}$$

Thus the first EM junction condition, in terms of intrinsic components, is

$$[A_a] = 0. \tag{45}$$

This can also be written as

$$[A_\alpha] = fk_\alpha, \tag{46}$$

with arbitrary scalar function f.

As in the non-null case $[A_a] = 0$ directly implies $[F_{ab}] = 0$, but it also follows from the equation (25)

$$F_{[\alpha\beta;\gamma]} = [F_{\alpha\beta}]k_\gamma + [F_{\beta\gamma}]k_\alpha + [F_{\gamma\alpha}]k_\beta = 0.$$

Due to antisymmetry, the only nontrivial contractions give

$$F_{[\alpha\beta;\gamma]}e^\alpha_a e^\beta_b N^\gamma = -[F_{ab}]$$
$$[F_{ab}] = 0.$$

So there is no other constraint than the one that follows from the first EM junction condition.

4.2 Second condition

Let's now look at the surface current term arising from

$$F^{\alpha\beta}{}_{;\beta} = \Theta(\tau)F^{+\alpha\beta}{}_{;\beta} + \Theta(-\tau)F^{-\alpha\beta}{}_{;\beta} - (-k_\mu u^\mu)^{-1}[F^{\alpha\beta}]k_\beta\delta(\tau).$$

Maxwell's equation gives

$$I_\alpha = C^{-1}[F_{\alpha\beta}]k^\beta, \tag{47}$$

with $C = -(-k_\mu u^\mu)4\pi$. The contractions with the basis vectors give:

$$\begin{aligned} e^\alpha_A: &\quad [F_{A\beta}k^\beta] = C\, I_A, \\ k^\alpha: &\quad [F_{\alpha\beta}k^\alpha k^\beta] = C\, I_\alpha k^\alpha = 0, \\ N^\alpha: &\quad [F_{\alpha\beta}N^\alpha k^\beta] = C\, I_\alpha N^\alpha. \end{aligned}$$

Due to $[F_{ab}] = 0$, we get $I_A = 0$, so the only remaining condition is

$$I_k = C^{-1}[\bar{F}], \tag{48}$$

with

$$I_\alpha N^\alpha = I_k, \qquad F_{\alpha\beta}N^\alpha k^\beta = \bar{F}.$$

This is the second EM junction condition. The surface four current has the form:

$$I_\alpha = -I_k k_\alpha \Rightarrow I^\alpha = -I_k k^\alpha. \tag{49}$$

Summary

The junction conditions using the intrinsic quantities of the null Σ are

$$[A_a] = 0 \tag{50}$$

$$4\pi I_k = -(-k_\mu u^\mu)^{-1}[F_{\alpha\beta}]N^\alpha k^\beta \tag{51}$$

5 A star with a surface charge density

As an example, we present a calculation of junction conditions for a star that has a uniform charge density at the surface while the rest of the star is electrically neutral.

Let us consider the following situation, as depicted in figure 1. We have an exterior Reissner-Nordström (RN) spacetime $\mathcal{V}^+$, joined to an interior spacetime $\mathcal{V}^-$, representing a star with constant density $\rho_\star$ and total charge Q, which is entirely confined to its surface. The hypersurface Σ separating these regions is a time-like hypersurface. The radial coordinate r is an affine parameter that is smooth across the boundary and the same on both sides. The coordinates on Σ can be straightforwardly chosen to be $y^a = (t, \theta, \phi)$.

We assume an idealized star that is static with constant density $\rho_\star$ and is spherically symmetric. The following metric in Schwarzschild coordinates describes the inner part of

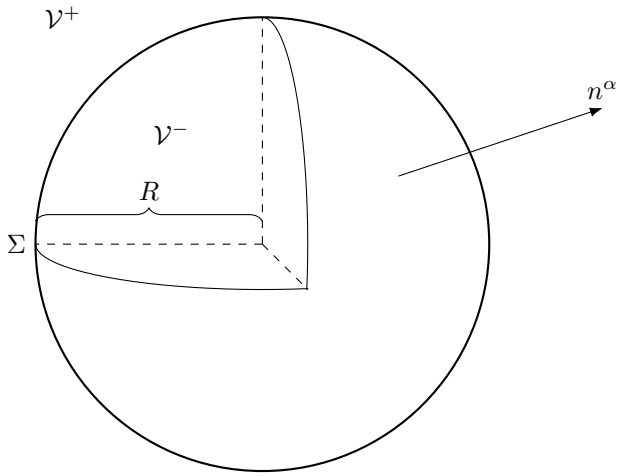


Figure 1: Visual representation of the star: manifolds $\mathcal{V}^-$ and $\mathcal{V}^+$ and the boundary Σ between them. The vector n^α is the normal vector that points to the spacetime $\mathcal{V}^+$ from the hypersurface Σ.

this star

$$ds^2 = -\left[\frac{3}{2}\left(1-\frac{2M_\star}{R}\right)^{\frac{1}{2}} - \frac{1}{2}\left(1-\frac{2M_\star r^2}{R^3}\right)^{\frac{1}{2}}\right]^2 dt^2 + \left(1-\frac{2m(r)}{r}\right)^{-1} dr^2 + r^2 d\Omega^2, \quad (52)$$

where $m(r) = \frac{4\pi}{3}\rho_\star r^3$ is the mass enclosed within radius r, R is the radius of the star, and $M_\star = m(R)$ is the total mass contributed by the density of the star [11]. Outside the star, in $\mathcal{V}^+$, the metric is RN

$$ds^2 = -f_{RN}dt^2 + f_{RN}^{-1}dr^2 + r^2 d\Omega^2, \quad (53)$$

with $f_{RN} = 1 - \frac{2M}{r} + \frac{Q^2}{r^2}$, where M is the total gravitational mass distinct from $M_\star$ due to the contribution of the total charge Q to the spacetime curvature.

Let us first calculate the Israel junction conditions to see the conditions for joining the spacetimes. We must first compute the induced metric and the extrinsic curvature to calculate the Israel junction conditions. The induced metric is given by

$$h_{ab} = g_{\alpha\beta} e_a^\alpha e_b^\beta, \quad (54)$$

which means that the metrics on the two sides of Σ (at $r = R$) are:

$$ds^2|_{\Sigma-} = -\left(1-\frac{2M_\star}{R}\right)dt^2 + R^2 d\Omega^2\,, \quad (55)$$

$$ds^2|_{\Sigma+} = -f_{RN}dt^2 + R^2 d\Omega^2\,. \quad (56)$$

The first Israel junction condition $[h_{ab}] = 0$ gives us

$$M = M_\star + \frac{1}{2}\frac{Q^2}{R}. \quad (57)$$

This relation shows that the total mass has a charge contribution in addition to the stellar mass $M_\star = \frac{4\pi}{3}\rho_\star R^3$, and this contribution is singular just like the charge distribution itself.

Using equation (8) we calculate the extrinsic curvature to be

$$K^-_{tt} = -\frac{M_\star}{R^2\sqrt{g^-_{RR}}} \qquad K^+_{tt} = \frac{1}{R^2\sqrt{g^+_{RR}}}\left(-M + \frac{Q^2}{R}\right) \tag{58}$$

$$K^-_{\theta\theta} = \frac{R}{\sqrt{g^-_{RR}}} \qquad K^+_{\theta\theta} = \frac{R}{\sqrt{g^+_{RR}}} \tag{59}$$

$$K^-_{\phi\phi} = \frac{R\sin^2\theta}{\sqrt{g^-_{RR}}} \qquad K^+_{\phi\phi} = \frac{R\sin^2\theta}{\sqrt{g^+_{RR}}} \quad , \tag{60}$$

where $g^\pm_{RR} \equiv g^\pm_{rr}|_{r=R}$. We see that the only discontinuous component is K_{tt}, which using (57) simplifies to

$$[K_{tt}] = \frac{1}{2R^2\sqrt{g_{RR}}}\left(\frac{Q^2}{R}\right), \tag{61}$$

where $\sqrt{g_{RR}} = \sqrt{g^\pm_{RR}}$. Thus the second junction condition is violated and there is a thin shell at Σ.

To see what contributes to this thin shell, let us calculate the energy-momentum tensor of Σ. Using the Lanczos equation (19) and (61) we get the components of S_{ab}:

$$S_{tt} = 0 \tag{62}$$

$$S_{\theta\theta} = -\frac{1}{16\pi}\sqrt{g_{RR}}\left(\frac{Q^2}{R}\right) \tag{63}$$

$$S_{\theta\theta} = -\frac{1}{16\pi}\sqrt{g_{RR}}\left(\frac{Q^2}{R}\right)\sin^2\theta \quad . \tag{64}$$

The singular charge density produces a singular energy-momentum tensor at Σ, not because of the mass of the charge carriers, which is included in the mass $M_\star$, but from the charge's contribution to the total mass of the system via equation (57). This equation tells that there is an extra layer effective of mass $M_\Sigma = \frac{1}{2}\frac{Q^2}{R}$ at Σ. This is also evident from the Q^2 dependence of the singular part of the energy-momentum tensor. It is an interesting result that while the electromagnetic fields themselves cannot contribute to the singular part of the energy-momentum tensor, the charge density does so via its effect on the spacetime curvature.

Let us now use the previously derived electromagnetic junction conditions. In $\mathcal{V}^+$ we have the stress tensor $F^{\alpha\beta}$ with the non-zero components $F^{tr} = \frac{Q}{r^2} = -F^{rt}$. In $\mathcal{V}^-$, where there is no electromagnetic field we have $F^{\alpha\beta} = 0$ The first electromagnetic junction condition is trivial because we can set the potential in V^- to be constant such that $A^-_\alpha(r) = A^+_\alpha(R)$. The third electromagnetic junction condition is also trivial since the only non-zero components of the electromagnetic tensor in $\mathcal{V}^+$ are $F^{tr} = -F^{rt}$ and r is the index along the normal, which means $F^+_{ab} = 0 \Rightarrow [F_{ab}] = 0$. The second electromagnetic junction condition is not

trivial since we have a star with a surface charge density:

$$\varepsilon[F^{a\beta}]n_\beta = 4\pi I^a \tag{65}$$

$$\Rightarrow \frac{Q}{R^2}\sqrt{g_{RR}} = 4\pi I^t\,. \tag{66}$$

This means that the charge density observed by a stationary observer, whose four-velocity is $u_\alpha = \sqrt{-g_{tt}}\delta_{t\alpha}$, is

$$I^t u_t = \frac{Q}{4\pi R^2}\sqrt{g_{RR}}\sqrt{-g_{tt}} = \frac{Q}{4\pi R^2}\,. \tag{67}$$

This is exactly the surface charge density we had assigned the surface. Thus, a star with a surface charge density satisfies the electromagnetic junction conditions.

6 Discussion

We have derived coordinate-independent electromagnetic (EM) junction conditions for both null and non-null hypersurfaces using distributional methods. The EM junction conditions for non-null hypersurfaces are quite straightforward and make sense when comparing to the classic jump conditions in electrodynamics. For the null case, the EM junction conditions require more careful analysis due to the nature of null manifolds, but they are also quite simple. Only the equation for the current density is surprising in the sense that it only has a component along the null generators of the hypersurface, unlike the non-null case where the current density could have components along all tangential directions of the hypersurface.

Usually the EM junction conditions are matched on a problem to problem basis starting from Maxwell's equations as it is quite straightforward to do so. While there is a general framework to calculate any junction conditions by Rascskó [9] there is no explicit derivation for the electromagnetic junction conditions in general relativity other than Rawson-Harris [8]. The purpose of this paper is to spell out the general conditions for any problem involving joining electromagnetic fields across a hypersurface.

Our derivation is new compared to the derivation of Rawson-Harris [8], which only considered non-null hypersurfaces. In addition to null hypersurfaces, our solution does not require the electromagnetic sources to be independent of time and works for dynamic situations. The only advantage of Rawson-Harris's approach is that it works with hypersurfaces that can arbitrarily change signature between timelike and spacelike.

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